

Food Insecurity as a Predictor of Diabetes and Obesity: An Ecological Analysis of U.S. Counties

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Abstract. Food insecurity is a major risk factor for chronic diseases and mental health challenges. To date, most studies on the relationship between food insecurity and obesity and diabetes prevalence have been performed on an individual level. The present study uses data from Feeding America’s Map the Meal Gap study, the American Community Survey, and Centers for Disease Control PLACES: Local Data for Better Health to examine relationships between rates of county-level food insecurity and diagnosed diabetes and obesity among adults. Descriptive analyses included generating correlation matrices, scatter plots, and choropleth maps to visualize spatial patterns and relationships among the variables. We conducted hierarchical linear regressions to examine associations between food insecurity and health outcomes, controlling for socioeconomic indicators and state-level factors. Additionally, random forest models were used to classify counties with diabetes and obesity rates above the national average. These three analytical approaches enabled a comprehensive examination of the relationship between food insecurity and the health outcomes of interest. Food insecurity was strongly associated with and highly predictive of both diabetes and obesity, with a stronger relationship observed for diabetes. Socioeconomic factors, including county racial composition, age, educational attainment, and median income were also explanatory of county-level variation in diabetes and obesity rates. These findings highlight the importance of addressing food insecurity to improve population-level health outcomes, especially in counties with a higher burden of diabetes and obesity rates.

Keywords: Food insecurity · Obesity · Diabetes.

1 Introduction

Food insecurity affects more than 47 million Americans each year (Dewey et al., 2025) [1]. Defined as “the limited or uncertain availability of nutritionally adequate and safe foods, or limited or uncertain ability to acquire acceptable foods in socially acceptable ways,” food insecurity is a major risk factor for chronic diseases and mental health and well-being challenges. Coping strategies employed by food insecure individuals such as choosing cheaper, nutrient poor foods or compensatory eating during times of greater food security have led to the paradoxical relationship between food insecurity and diet-related chronic diseases like diabetes and obesity.

There is a wealth of literature documenting the connections between food insecurity and obesity and diabetes outcomes. Longitudinal studies have shown greater odds of obesity among children when exposed to persistent food insecurity (Metallinos-Katsara et al., 2012 [2]; Nikolaus et al., 2022 [3]) and others have documented that food insecurity was associated with a greater increase in BMI among people of all ages (Cheung et al., 2015 [4]; Lohman et al., 2018 [5]; Nikolaus et al., 2022 [3]). Studies using National Health and Nutrition Examination Survey (NHANES) data have found

associations between individual-level food insecurity and both subject-reported and laboratory examination diabetes outcomes (Seligman et al., 2007 [6]; Seligman et al., 2010 [7]).

To date, most studies on the relationship between food insecurity and obesity and diabetes prevalence have been performed on an individual level primarily using data from the NHANES study. There has been one county-level study that found food insecurity was the eleventh strongest predictor of county-level food insecurity outcomes (Salerno et al., 2024 [8]). The present study intends to perform an ecological analysis and document the population-level relationships between rates of food insecurity and diet-related chronic diseases. Specifically, we examine whether county-level food insecurity rates are associated with county-level prevalence of diabetes and obesity controlling for other county socioeconomic data such as income and education attainment. This work is important because public health policies are often implemented at the county and local level and this work can guide resource allocation to target communities with the greatest need.

2 Data and Methods

2.1 Data sources

All data were collected at the county level. Our analysis integrates three primary sources:

- **Feeding America: Map the Meal Gap (MMG, 2010–2022):** This dataset is the primary source for our variables related to food insecurity and economic distress. It provides granular estimates of overall and child food insecurity for every county in the United States. It spans from 2010 to 2022, allowing us to model both short-term and long-term trends in food insecurity.
- **Centers for Disease Control and Prevention (CDC) PLACES (2022):** This dataset provides county-level estimates of various adult health indicators for U.S. counties. It includes over 29 measures across categories including health outcomes, risky behaviors, disabilities, etc. For our analysis, we examine obesity and diagnosed diabetes rates among adults and consider these as population-level disease burden in 2022.
- **American Community Survey (ACS, 2022):** ACS provides yearly statistics on U.S. communities' economic, social, and demographic characteristics. We extracted information including income, education, and age distribution data as these are often associated with public health outcomes.

2.2 Variable descriptions

- **Food insecurity indicators:** Average food insecurity rates over three time intervals for different lengths of exposure to food insecurity: long-term (2010-2022), medium-term (2015-2019), short-term (2020-2022). Cumulative food insecurity rates were also calculated. We also considered Cost per meal and Annual weighted shortfall (later was transformed to Shortfall Per Capita). Throughout the semester, our analysis evolved as we clarified what makes a meaningful health indicator.
- **Health outcomes:** Diabetes prevalence and obesity prevalence are the key dependent variables. These are expressed as percentages of the adult population diagnosed with the respective condition in 2022.
- **Socioeconomic factors:** Median household income captures economic resources at the county level. Educational attainment is represented as the proportion of adults with a high school education. Racial composition and age distribution provide demographic context for our analyses and enable adjustment for potential confounding variables in regression and predictive models.

2.3 Data pre-processing and feature engineering

Because the MMG data spans thirteen years and is released as separate annual excel files, we first converted these files to comma-separated values (.CSV) format and merged them into a single longitudinal dataset. Inevitably, we found some inconsistencies in column names and data types. The same variable frequently appeared with different names depending on the year as a prefix or suffix (for example: “2014 Weighted Shortfall” or “# of Food Insecure Persons in 2012”). We used Regular Expression matching to strip out the embedded year and mapped synonymous fields to a previously defined column name (using 2012 as our baseline). Federal Information Processing Standard (FIPS) codes, the unique identifier for each county, was stored in a mix of integer, float, and string formats across files. We resolved these discrepancies by converting all FIPS fields to string-based formats and adding “0” to the left side if needed. Percentage values were converted to 0 – 1 floating values. The goal was to create a clean, longitudinal MMG dataset suitable for merging with ACS and CDC PLACES.

Once the MMG data were merged, we created the following indicators relevant to our analyses:

- **Shortfall Per Capita:** This was calculated by dividing the total annual food budget shortfall by the county’s estimated total population. This standardizes the weighted shortfall so we can fairly compare counties of different population sizes.
- **Short-, medium-, and long-term exposure measures and their cumulatives:** We constructed three distinct measures based on non-overlapping time windows of annual food rates. Short-, medium-, and long-term exposure measures are the mean food insecurity of 2020 – 2022 (recent), 2015–2019 (5 years earlier) and (2010–2014) (5 years earliest) respectively. The medium- and long-term cumulative exposure measures are calculated over 2015–2022 and 2010–2022 time periods, respectively. We created these time lags for two reasons: (1) to establish that the exposure comes before the outcome, arguing that food insecurity might cause health problems rather than the reverse; (2) to initiate plausible time period for sustained food insecurity to cause measurable health problems.

Changes in FIPS codes throughout the time period of the MMG dataset was the biggest issue we encountered in processing the data. This issue was most prominent in Connecticut as the counties were officially changed to census planning regions in 2022 with different boundaries and FIPS codes between 2010–2022. There were additional isolated incidents of counties that merged together or separated and obtained new FIPS codes. Food insecurity rates for these 13 counties (0.41% of all U.S. counties) in years before 2022 required imputation. For the sake of time on this project, we removed these counties from our analyses since they did not comprise a large percentage of our population.

The CDC PLACES dataset was structurally cleaner, but still needed to be transformed from long to wide format. In addition, we had to filter out a large set of health indicators that were not required for our analysis. We also ensured the FIPS codes followed the same five-character format as the MMG data.

The ACS datasets were also structurally clean and already existed in a wide format. In order to include all of the socioeconomic variables of interest, we merged three ACS datasets from 2022 based on FIPS codes and filtered for our required variables. One consideration with the ACS data is occasionally small counties (population $\leq 65,000$) are excluded, which can lead to scattered missing values. We did not impute these values as this would introduce noise into socioeconomic predictors.

After separately processing each of the three data sources, we merged the three files into a singular county-level file using standardized FIPS codes. We validated the merge by checking for duplicates, assessing missingness, and verifying data consistency across the years.

2.4 Methods and Models

Our primary focus was to analyze the correlation between our food insecurity variables and diabetes and obesity prevalence. We began with the correlation matrix to quantify the strength and direction of the linear relationship between pairs of variables. In our setup, this served as a first step to see if food insecurity is associated with both conditions. We chose this method for its interpretability in highlighting initial linear associations and to guide further explanatory analysis.

Next, we developed a series of Ordinary Least-Squares (OLS) regression models to quantify the association between food insecurity and health outcomes. The three model specifications were:

- **Unadjusted model:** We first investigated the direct link between FI rate and Diabetes and Obesity prevalence. This provided the unadjusted association before accounting for any other factors.
- **SES-adjusted models:** We added our socioeconomic and demographic control variables (income, age, education, racial composition and unemployment). The addition of these variables enabled us to isolate the specific association of food insecurity when holding county sociodemographic characteristics constant.
- **State fixed effects model:** To make our model more robust, we introduced controls for every U.S. state. This accounts for unobserved factors such as state policies and healthcare access environments. As a result, our model can effectively compare counties with others in the same state while minimizing bias caused by geographic regions.

We compared the R^2 values across our sequential models. Controls are added provide insight into the relative contribution of food insecurity, socioeconomic structure, and geographic context in explaining variation in health outcomes..

We approached the predictive analysis from a classification perspective. We wanted to predict whether a county's diabetes (or obesity) rate exceeded the national average. We used a Random Forest classifier because of its flexibility, robustness to nonlinearities, and interpretability. Model performance was evaluated using accuracy, precision, recall, and confusion matrices. To improve generalization and avoid overfitting, model hyperparameters are selected using a cross-validated grid search. We tune key parameters and select the configuration that performs best. Given the public health implications, we considered the cost of false positives (identifying a high-burden county when it is not) and false negatives (failing to identify a county that exceeds the national diabetes rate). Overlooking high-burden counties can lead to under-allocation of available resources.

3 Analysis Results

Our analysis proceeded in three stages: descriptive analysis, explanatory analysis, and predictive analysis.

3.1 Descriptive analysis

For the descriptive analysis, we wanted to visualize the geographic spread of variables and establish relationships between predictors and outcomes. When plotting the Prevalence of Diabetes/Obesity and the Shortfall Per Capita, we saw that the counties with high rates of Diabetes/Obesity burden often overlapped with the same regions as those with high food deficits.

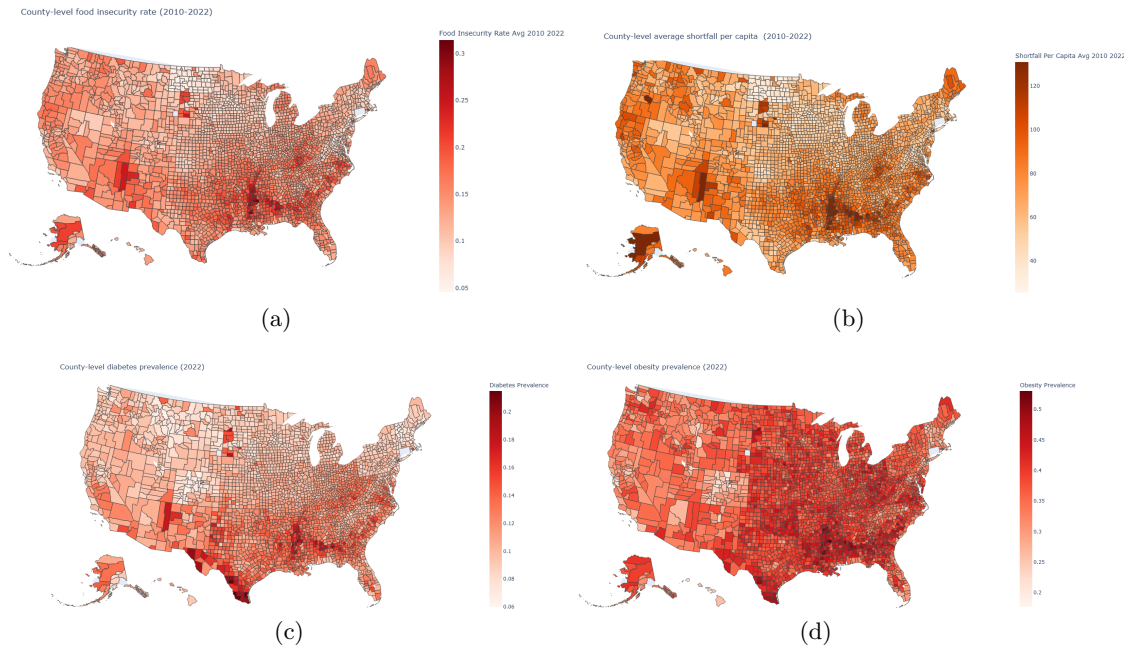


Fig. 1. (a) County-level average food insecurity rate (2010-2022) (b) County-level average shortfall per capita (2010-2022) (c) County-level diabetes prevalence in 2022 (d) County-level obesity prevalence in 2022

Counties with high average food insecurity rate and average shortfall per capita overlap geographically with counties having high diabetes and obesity prevalence.

We observed a pattern called the "diabetes belt," [9] with some of the highest rates concentrated in the Southeast. Parts of Texas and Arizona also experienced high rates of diabetes. Some counties in these regions have diabetes rates exceeding 14 percent. Obesity prevalence follows a similar geographic pattern, with many counties having more than 35 percent of adults classified as obese. This visual evidence provided initial support for our hypothesis that food insecurity may be associated with worse metabolic health outcomes.

To demonstrate the direction of the relationship, we created a scatterplot of diabetes and obesity prevalence (Figure. 2) with each point representing one county. Darker points tend to cluster toward the upper right corner of the plot, meaning counties with higher food insecurity rates generally have both higher diabetes rates and higher obesity rates. This suggests that food insecurity is associated with both conditions simultaneously.

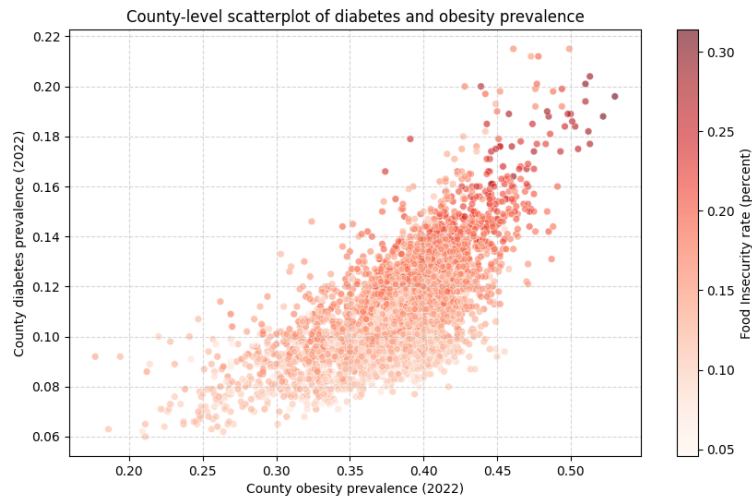


Fig. 2. County-level scatterplot of diabetes and obesity prevalence. Darker points (higher FI rate) tend to be on the higher end of diabetes and obesity prevalence percentages

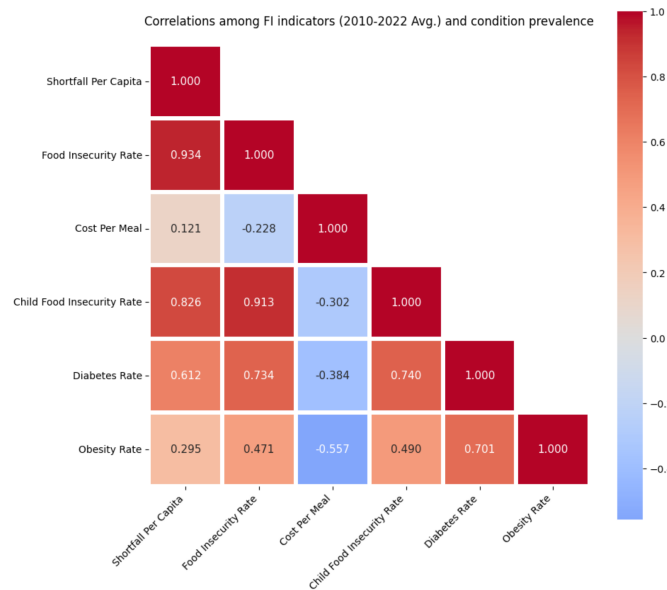


Fig. 3. Heatmap of correlation matrix between FI indicators (Avg. 2010-2022) and condition. Cost per meal have negative correlation both with FI indicators and condition outcomes.

We computed a correlation matrix for the main food insecurity indicators and the two health outcomes (Figure 3.). The main food insecurity indicators (overall food insecurity rate, shortfall per capita, and child food insecurity rate) are extremely strongly correlated with one another (all above 0.8). This makes sense conceptually as all measure food insecurity from different angles. These indi-

cators also show moderately strong, positive correlations with both diabetes and obesity prevalence (notably food insecurity rate and diabetes rate are positively correlated at 0.734). An unexpected finding is that cost per meal is negatively correlated with both food insecurity indicators and health outcomes. Counties with higher meal costs tend to have lower food insecurity rates (correlation of -0.228) and lower diabetes prevalence (correlation of -0.384) and obesity prevalence (-0.557). This is likely due to deeper structural differences between high cost of living counties and low cost of living counties. Counties with higher meal costs tend to be more urban and have better access to health services, which may reduce chronic disease rates. Whereas counties with lower meal costs are often rural and with less access to healthcare which can contribute to higher rates of chronic diseases. This prompts the importance of controlling for socioeconomic factors in our explanatory analysis.

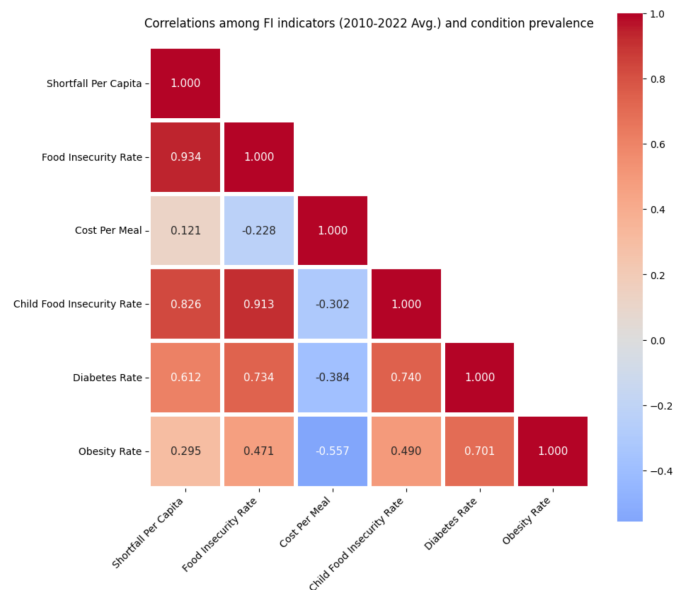


Fig. 4. Heatmap of correlation matrix between FI rates (with exposure components) and condition prevalence.

We also examined correlations using our constructed short-term, medium-term, and long-term food insecurity exposure measures (Figure 4.). As expected, all exposure measures show strong positive correlations with diabetes diagnoses (ranging from 0.688 to 0.734) and positive correlations with obesity prevalence (0.407 to 0.496). These patterns suggest that chronic, sustained exposure to food insecurity captures deeper structural vulnerability than more recent exposure. For this reason, we used the long-term cumulative food insecurity indicators in our subsequent analyses.

3.2 Explanatory analysis

To examine how food insecurity relates to county-level diabetes and obesity prevalence, we performed a sequence of hierarchical Ordinary Least Squares (OLS) regression models.

Model 1 (Food Insecurity Rate only) OLS Regression Model

We began with a baseline specification that includes only the mean food insecurity rate from 2010 to 2022 as our predictor. Regression results from Model 1 (Food insecurity predictor only) show that mean 2010-2022 food insecurity rate is significantly and positively associated with diabetes rates ($\beta = 0.465$, $p < 0.001$) and obesity rates ($\beta = 0.602$, $p < 0.001$). The model accounts for 53.8% of the variance in diabetes rates and 22.2% of the variance in obesity rates (Table 1.; Figure 5.)

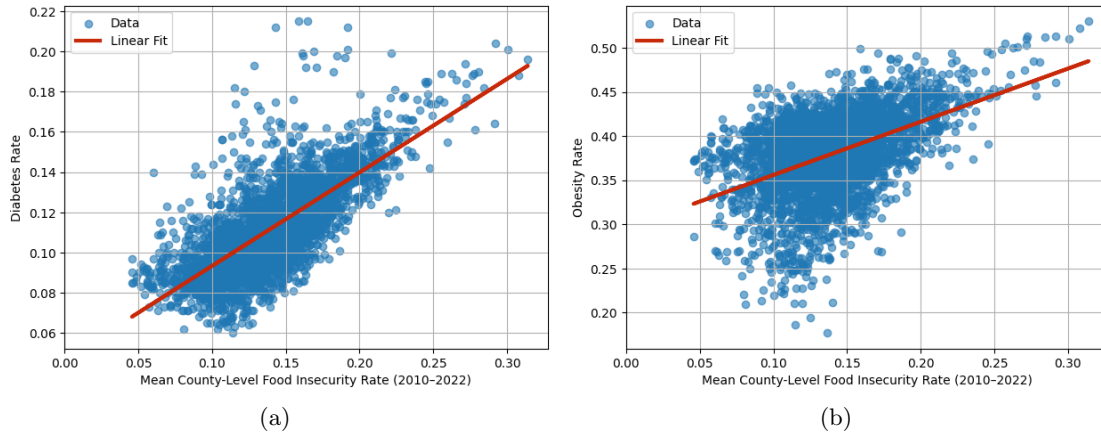


Fig. 5. (a) OLS Regression of county-level Mean Food Insecurity Rate (2010-2022) and Diabetes Rate (b) OLS Regression of county level Mean Food Insecurity Rate (2010-2022) and Obesity Rate

Table 1. Model 1 (Food Insecurity Rate only) OLS Regression Model

Model 1a: Diabetes Rate					Model 1b: Obesity Rate				
Variable	Coef	SE	t	p	Variable	Coef	SE	t	p
Constant	0.0468	0.0010	42.289	0.000	Constant	0.2957	0.0030	102.203	0.000
Mean FI rate 2010-22	0.4652	0.0077	60.306	0.000	Mean FI Rate 2010-22	0.6022	0.0202	29.853	0.000
Model Statistics					Model Statistics				
R-squared	0.538				R-squared	0.222			
Adj. R-squared	0.537				Adj. R-squared	0.221			
F-statistic	3637				F-statistic	891.2			
Prob(F)	0.000				Prob(F)	1.58e-172			
N	3130				N	3130			
Log-Likelihood	8574.5				Log-Likelihood	5566.0			
Durbin-Watson	1.147				Durbin-Watson	0.950			
Omnibus	648.251				Omnibus	301.651			
Prob(Omnibus)	0.000				Prob(Omnibus)	0.000			
Jarque-Bera	2275.905				Jarque-Bera	417.899			
Prob(JB)	0.000				Prob(JB)	1.80e-91			
Kurtosis	6.658				Kurtosis	3.916			
Cond. No.	28.1				Cond. No.	28.1			

Model 2 (Food Insecurity Rate + Socioeconomic Predictors) OLS Regression Model

In this model, we add key socioeconomic and demographic controls, including median household income, educational attainment, racial composition, and age structure. These are confounding factors that can affect the relationship between food insecurity and health. Median income measures disadvantage that could confound the relationship between food insecurity and chronic disease. Education affects health in multiple ways, including access to information and health literacy. We include race and age because health disparities and food insecurity rates differ by these demographics.

Table 2. Model 2 (Food Insecurity Rate + Socioeconomic Predictors) OLS Regression Model

Model 2a: Diabetes Rate					Model 2b: Obesity Rate				
Variable	Coef	SE	t	p	Variable	Coef	SE	t	p
Constant	0.2927	0.0040	68.489	0.000	Constant	0.6474	0.014	46.243	0.000
Mean FI Rate 2010-22	0.1406	0.0091	15.453	0.000	Mean FI Rate 2010-22	-0.0696	0.030	-2.334	0.020
Med. HH Income	-3.4e-07	1.9e-08	-17.7	0.000	Med. HH Income	-1.684e-06	6.23e-08	-27.033	0.000
Per with HS Diploma	-0.1607	0.0043	-37.466	0.000	Per with HS Diploma	-0.1235	0.014	-8.790	0.000
Per White	-0.0342	0.0015	-23.009	0.000	Per White	-0.0096	0.005	-1.983	0.047
Per > 65 years old	-0.0532	0.0046	-11.610	0.000	Per > 65 years old	-0.1788	0.015	-11.909	0.000
Model Statistics					Model Statistics				
R-squared	0.786				R-squared	0.434			
Adjusted R-squared	0.786				Adjusted R-squared	0.433			
F-statistic	2293				F-statistic	479.3			
Prob(F-statistic)	0.000				Prob(F-statistic)	0.000			
No. Observations	3130				No. Observations	3130			
Log-Likelihood	9779.1				Log-Likelihood	6064.7			
Durbin-Watson	0.931				Durbin-Watson	1.030			
Omnibus	48.248				Omnibus	141.326			
Prob(Omnibus)	0.000				Prob(Omnibus)	0.000			
Jarque-Bera	71.666				Jarque-Bera	169.616			
Prob(JB)	0.000				Prob(JB)	1.47e-37			
Skew	0.164				Skew	-0.491			
Kurtosis	3.664				Kurtosis	3.581			
Cond. No.	3.33e+06				Cond. No.	3.33e+06			

Regression results from Model 2 (Food Insecurity Rate and socioeconomic predictors) show that mean 2010-2022 food insecurity rate remains significantly and positively associated with diabetes rates ($\beta = 0.1406$, $p < 0.001$) and significantly negatively associated with obesity rates ($\beta = -0.0696$, $p < 0.05$). The coefficients for each of the socioeconomic predictors are shown below in Table 2. Model 2 accounts for 78.6% of the variance in diabetes rates and 43.4% of the variance in obesity rates (Table 2.).

The association between food insecurity rates and diabetes and obesity rates decreases when socioeconomic predictors are added to the model, suggesting that some of the effect of food insecurity in the previous model may be explained by median household income, education attainment, and various demographic characteristics. Interestingly, in Model 2, the direction of the relationship between food insecurity rates and obesity rates reverses direction (i.e., negative coefficient).

Model 3 (Food Insecurity Rate + Socioeconomic Predictors + State Fixed Effects) OLS Regression Models

Regression results from Model 3 (Food Insecurity Rate + Socioeconomic Predictors + State Fixed Effects) show that mean 2010-2022 food insecurity rate remains significantly and positively associated with diabetes ($\beta = 0.0589$, $p < 0.001$) and obesity rates ($\beta = 0.1202$, $p < 0.001$). The coefficients for each of the socioeconomic predictors are shown below. Model 3 accounts for 92.3% of the variance in diabetes rates and 75.0% of the variance in obesity rates (Table 3.)

The addition of the state fixed effects further reduced the magnitude of the effect of food insecurity on diabetes and obesity rates. This finding indicates that unobserved state-level variables contribute to the variation in health outcomes. When controlling for state-level factors in addition to socioeconomic factors, the direction of the relationship between food insecurity and obesity becomes positive again.

Table 3. Model 3 (Food Insecurity Rate + Socioeconomic Predictors + State Fixed Effects) OLS Regression Models (State coefficients omitted)

Model 3a: Diabetes Rate					Model 3b: Obesity Rate				
Variable	Coef	SE	t	p	Variable	Coef	SE	t	p
Intercept	0.2529	0.003	79.763	0.000	Intercept	0.5618	0.011	48.852	0.000
Mean FI Rate 2010-22	0.0589	0.008	7.617	0.000	Mean FI Rate 2010-22	0.1202	0.028	4.282	0.000
Med. HH Inc.	-3.758e-07	1.4e-08	-26.761	0.000	Med. HH Inc.	-1.039e-06	5.09e-08	-20.407	0.000
Per with HS Diploma	-0.1124	0.003	-37.591	0.000	Per with HS Diploma	-0.1423	0.011	-13.111	0.000
Per White	-0.0562	0.001	-46.716	0.000	Per White	-0.0421	0.004	-9.646	0.000
Per > 65 years old	-0.0183	0.003	-6.011	0.000	Per > 65 years old	-0.0527	0.011	-4.787	0.000
Model Statistics					Model Statistics				
R-squared	0.923				R-squared	0.750			
Adjusted R-squared	0.921				Adjusted R-squared	0.746			
F-statistic	680.7				F-statistic	170.8			
Prob(F-statistic)	0.000				Prob(F-statistic)	0.000			
No. Observations	3130				No. Observations	3130			
Log-Likelihood	11376				Log-Likelihood	7342.7			
Durbin-Watson	2.012				Durbin-Watson	2.019			
Omnibus	471.915				Omnibus	119.278			
Prob(Omnibus)	0.000				Prob(Omnibus)	0.000			
Jarque-Bera	2185.587				Jarque-Bera	242.250			
Prob(JB)	0.000				Prob(JB)	2.49e-53			
Skew	0.651				Skew	-0.261			
Kurtosis	6.881				Kurtosis	4.259			
Cond. No.	5.24e+06				Cond. No.	5.24e+06			

3.3 Predictive analysis

We created a binary classification task that asks whether we can predict a county has diabetes and obesity rates above or below the national average. We used Random Forest classification with predictors including mean food insecurity rate for 2010 – 2022, median household income, age, race, and high school education attainment. To ensure reliable measure of performance, we utilized 5-fold stratified cross-validation (CV).

The model demonstrated consistent performance. Table 4. summarizes the mean performance metrics, calculated across all 5 CV folds.

Table 4. Random Forest Classifier: Mean performance metrics (5-fold CV)

Metric (Mean $\pm$ Std)	Diabetes Model	Obesity Model
Accuracy	86.42 $\pm$ 1.33	72.84 $\pm$ 1.07
Precision	85.61 $\pm$ 1.92	75.31 $\pm$ 0.38
Recall	83.67 $\pm$ 2.59	75.56 $\pm$ 2.95
F1-Score	84.60 $\pm$ 1.63	75.40 $\pm$ 1.46
AUC	93.93 $\pm$ 0.66	81.19 $\pm$ 0.72

The performance clearly shows that the features developed in this study are significantly more predictive of diabetes prevalence than obesity prevalence. The diabetes model achieved high performance across all metrics, with an accuracy score of 86.42% and AUC score of 93.93%. Our predictors, including food insecurity rate and socioeconomic factors, strongly provide information to distinguish high-risk diabetes counties. Precision and recall scores are 85.61% and 83.67% respectively. The model successfully identified the majority of truly high-risk counties. Based on these same predictors, the obesity model has a lower overall predictive capacity, yet still performs better than random guessing.

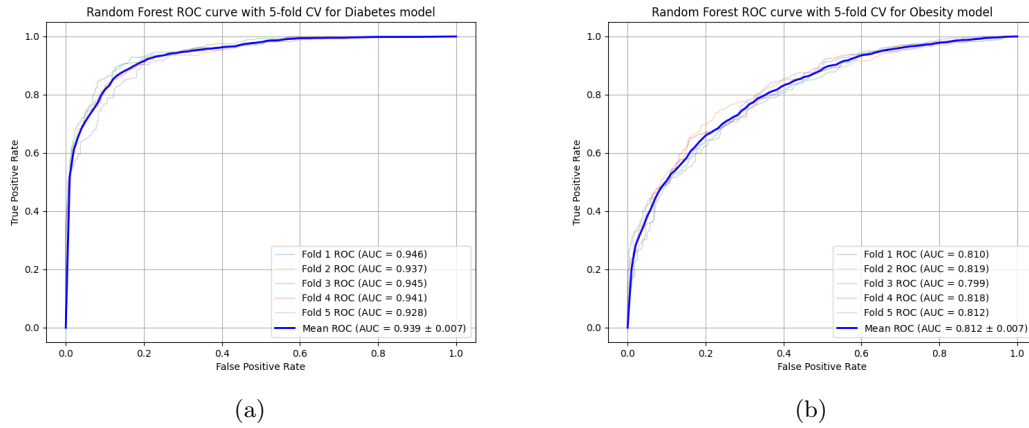


Fig. 6. (a) ROC curve with 5-fold CV for Diabetes model (b) ROC curve with 5-fold CV for Obesity model

Building upon this model, we performed 5-fold CV grid search (`GridSearchCV`) over a defined parameter space for maximum depth, minimum split, and number of trees in the forests. A total of 32 parameter combinations were evaluated, which resulted in 160 total model fits. The optimized hyperparameter settings for each model were:

- **Diabetes model:** `max_depth = 10`, `min_samples_split = 5`, `n_estimators = 400`.

- **Obesity model:** `max_depth = 10`, `min_samples_split = 2`, `n_estimators = 300`.

Utilizing these hyperparameters, the AUC for the diabetes and obesity models are 94.1% and 81.7%, respectively. The performance improved slightly across both models.

Random Forests also enable us to find which features matter most for prediction. The feature importance scores are summarized as the following:

Table 5. Random Forest Feature Importance Scores

Feature	Diabetes model	Obesity model
<code>per_hs_grad</code>	0.347	0.217
<code>fi_mean_10_22</code>	0.282	0.181
<code>median_household_income</code>	0.159	0.310
<code>percent_white_only</code>	0.122	0.133
<code>percent_over65</code>	0.090	0.159

For the diabetes model, education attainment (`per_hs_grad`) is the strongest predictor (approximately 0.346). This shows a very strong link between education and diabetes rate. This conceptually makes sense because managing diabetes and preventing its development requires certain health knowledge. Food insecurity rate (`fi_mean_10_22`) was the second most important predictor (around 0.281).

For the obesity model, median household income (`median_household_income`) was the most important predictor (0.310). Education (`per_hs_grad`) was the second most important (0.216). This pattern tells us that the model relies strongly on socioeconomic conditions to identify counties with high obesity rates. Higher income and greater educational attainment allow people to have better access to healthy food and utilize healthcare services. The food insecurity measure was the third most important factor. All in all, food insecurity, despite not being the strongest factor, adds crucial predictive power to our models that other socioeconomic factors can not capture.

4 Discussion

4.1 Key findings

Results from this study strongly support a positive association between food insecurity rates and health outcomes in U.S. counties. This study found that food insecurity exposure of any length is more strongly correlated with diabetes outcomes than obesity outcomes. This finding was further demonstrated by our explanatory analyses where food insecurity explained more of the variance in county-level diabetes outcomes than in obesity outcomes. High accuracy metrics and mean AUC scores from our predictive analyses further support this finding.

Our explanatory models also suggest that much of the variation in diabetes and obesity outcomes are explained by socioeconomic factors (i.e., demographics, education attainment, income, etc.) and unobserved state-level factors in addition to food insecurity rates. These unobserved state-level factors may include local policies that influence access to healthcare services.

Another major finding from our predictive analyses is education attainment was the strongest predictor of whether a county had a diabetes rate higher than the national average, but median household income was the strongest predictor of this relationship for obesity rates. The differences in feature importance between the diabetes and obesity models suggest different socioeconomic factors contribute to these two health outcomes in different ways. It may be that a high school level education facilitates increased access to healthcare that addresses diabetes and that income has a more direct influence on the types of food people can buy and their risk of obesity. Food insecurity rates were the second and third strongest predictor of whether a county had higher than national average diabetes and obesity rates, respectively.

4.2 Limitations

One key limitation of our study is its reliance on correlational analysis, which, by nature, cannot establish causation. While we identified significant associations between certain food insecurity indicators, there may be unmeasured confounding variables influencing these relationships. For example, counties with poor access to clinics or with low health insurance coverage might have higher disease rates. We couldn't capture the food density. Two counties with similar income metrics might have different health outcomes simply because one has lots of accessible grocery stores while fast-food chains and convenience stores are more accessible to the other one.

Furthermore, the generalizability of our findings is limited. As our analysis uses aggregated data at the county level, we are at risk of experiencing ecological fallacy. We can only draw conclusions about counties as a whole and cannot infer relationships at the individual level. Additionally, this study did not address spatial autocorrelation, where counties that are nearby one another are more similar to each other than counties that are far away from one another.

We want to acknowledge the time mismatch between our data sources. Our food insecurity measures span 2010-2022 while our control variables are from 2022 - the same time period of our health outcomes. Ideally, we would measure confounding factors over the years to properly control for factors influencing both food insecurity in the past and health outcomes now. We made assumptions that socioeconomic characteristics of counties change slowly, so 2022 values reasonably approximate earlier periods. This could be acceptable for racial and education components, yet less reliable for income and unemployment.

Our predictive models such as Random Forests capture associations but do not establish causality. Counties identified as "high risk" may be so due to unobserved structural factors mentioned above. Therefore, our predictive findings should be interpreted as indicators of vulnerability, rather than causation.

4.3 Future work

Future research should more closely examine the temporal relationship between food insecurity and obesity and diabetes. Obesity and diabetes diagnoses lag behind the changes in diet associated with food insecurity and less healthy eating. Time-series analyses could also explore how food insecurity rates track with socioeconomic factors over time such as median household income and educational attainment.

Our analyses showed that there are counties that have about average or above average food insecurity rates but remain well below the national average obesity rate. It would be important to examine these counties further to better understand the factors that enable these counties to fare better than those with similar food insecurity rates. It may be that there are policies in these counties that facilitate more adequate food access or access to healthcare.

Future work should also investigate whether the relationship between food insecurity and health outcomes found in this study remains if spatial autocorrelation is accounted for. This study attempts to account for unobserved state-level effects, but counties within the same state may be subject to autocorrelation as well.

4.4 Reflection

This project challenged us to perform a data science workflow from scratch. We learned that data collection and cleaning often take much more time than we expect. We learned that we need to document our progress and write reproducible code so that others can verify and reuse. We learned how to think about addressing food insecurity and health disparities. Food insecurity, though not the strongest predictor of health outcomes, still provide information for health consequences that other factors cannot capture. At the same time, we must acknowledge that food insecurity is deeply linked with poverty, racial demographics, lack of education, and other systemic inequities. Thus, addressing food insecurity requires us to address other root causes in economic and social inequality. We are inspired to work on future research projects with stronger study designs and better knowledge on this matter.

References

1. Dewey, A., Hilvers, J., Dawes, S., Harris, V., Hake, M., & Engelhard, E. (2025). Map the Meal Gap: A Report of Local Food Insecurity and Food Costs in the United States in 2023. Feeding America National Organization. <https://www.feedingamerica.org/research/map-the-meal-gap/overall-executive-summary> Published May 14, 2025
2. Metallinos-Katsaras, E., Must, A., & Gorman, K. (2012). A Longitudinal Study of Food Insecurity on Obesity in Preschool Children. *Journal of the Academy of Nutrition and Dietetics*, 112(12), 1949–1958. <https://doi.org/https://doi.org/10.1016/j.jand.2012.08.031>
3. Nikolaus, C. J., Hebert, L. E., Zamora-Kapoor, A., & Sinclair, K. I. (2022). Risk of Food Insecurity in Young Adulthood and Longitudinal Changes in Cardiometabolic Health: Evidence from the National Longitudinal Study of Adolescent to Adult Health. *The Journal of Nutrition*, 152(8), 1944–1952. <https://doi.org/https://doi.org/10.1093/jn/nxac055>
4. Cheung, H. C., Shen, A., Oo, S., Tilahun, H., Cohen, M. J., & Berkowitz, S. A. (2015). Food Insecurity and Body Mass Index: A Longitudinal Mixed Methods Study, Chelsea, Massachusetts, 2009–2013. *Preventing Chronic Disease*, 12. <https://doi.org/https://doi.org/10.5888/pcd12.150001>
5. Lohman, B. J., Neppl, T. K., Lee, Y., Diggs, O. N., & Russell, D. (2018). The Association between Household Food Insecurity and Body Mass Index: A Prospective Growth Curve Analysis. *The Journal of Pediatrics*, 202, 115–120.e1. <https://doi.org/https://doi.org/10.1016/j.jpeds.2018.05.052>
6. Seligman, H. K., Bindman, A. B., Vittinghoff, E., Kanaya, A. M., & Kushel, M. B. (2007). Food Insecurity is Associated with Diabetes Mellitus: Results from the National Health Examination and Nutrition Examination Survey (NHANES) 1999–2002. *Journal of General Internal Medicine*, 22(7), 1018–1023. <https://doi.org/https://doi.org/10.1007/s11606-007-0192-6>
7. Seligman, H. K., Laraia, B. A., & Kushel, M. B. (2010). Food Insecurity Is Associated with Chronic Disease among Low-Income NHANES Participants. *The Journal of Nutrition*, 140(2), 304–310. <https://doi.org/https://doi.org/10.3945/jn.109.112573>

8. Salerno, P. R. V. O., Qian, A., Dong, W., Deo, S., Nasir, K., Rajagopalan, S., & Al-Kindi, S. (2024). County-level socio-environmental factors and obesity prevalence in the United States. *Diabetes, Obesity and Metabolism*, 26(5), 1766–1774. <https://doi.org/https://doi.org/10.1111/dom.15488>
9. Barker, Lawrence E et al. “Geographic distribution of diagnosed diabetes in the U.S.: a diabetes belt.” *American journal of preventive medicine* vol. 40,4 (2011): 434-9. <https://doi.org/10.1016/j.amepre.2010.12.019>